

P2 Procurement & Test Pack

Quotation BOM, RFQ instructions, assembly, calibration, and acceptance

REV	A	LATITUDE	42.1 deg N	RELEASE	FOR QUOTATION / PHYSICAL TEST PENDING
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Release gate

Authorize coupon manufacture first. Release the production cams only after P2-120 through P2-122 pass. Do not accept machining from STL and do not treat software verification as physical acceptance.

P2 Bill of Materials

Status: Quotation BOM, Rev A Currency and supplier are intentionally open until the manufacturing package is reviewed.
Quantities include one assembly unless a spare is stated.

Custom CNC and cut parts

Item	Qty	Description	Material / process	Master files
P2-101	1	Daylight-duration cam	10 mm POM-C, 3-axis profile finish	STEP, DXF, CSV, PDF
P2-102	1	Sunrise-azimuth cam	10 mm POM-C, 3-axis profile finish	STEP, DXF, CSV, PDF
P2-103	2	Cam hub	6061-T6, turn/mill, clear anodize	STEP, DXF, PDF
P2-104	1	Cam module plate	6 mm 6061-T6, mill, clear anodize	STEP, DXF, PDF
P2-105	1	Daylight span lever	6 mm 6061-T6, mill, clear anodize	STEP, DXF, PDF
P2-106	1	Azimuth span lever	6 mm 6061-T6, mill, clear anodize	STEP, DXF, PDF
P2-107	1	Time carrier plate	6 mm 6061-T6, mill, clear anodize	STEP, DXF, PDF
P2-108	1	Time module plate	6 mm 6061-T6, mill, clear anodize	STEP, DXF, PDF
P2-109	1	Azimuth module plate	6 mm 6061-T6, mill, clear anodize	STEP, DXF, PDF
P2-110	1	Sunrise time shaft	303 stainless, turn/grind	STEP, PDF
P2-111	1	Sunset output shaft	303 stainless, turn/grind	STEP, PDF
P2-112	1	Azimuth inner shaft	303 stainless, turn/grind	STEP, PDF
P2-113	1	Azimuth outer tube	303 stainless, turn/grind	STEP, PDF
P2-115	1	Removable guard	4.5 mm clear polycarbonate, route	STEP, DXF, PDF
P2-116	2	Functional time dial	2.5 mm engraved/printed anodized aluminum	DXF, PDF
P2-117	1	Functional azimuth dial	same as P2-116	DXF, PDF
P2-118	2	Time hand	1.6 mm 5052-H32, laser/waterjet	STEP, DXF, PDF
P2-119	2	Azimuth hand	1.6 mm 5052-H32, laser/waterjet	STEP, DXF, PDF
P2-120	1	Constant-radius cam runout coupon	10 mm POM-C	STEP, DXF, PDF
P2-121	1	32 mm hub pilot fit coupon	10 mm POM-C	STEP, DXF, PDF
P2-122	1	100 mm lever span gauge	6 mm 6061-T6	STEP, DXF, PDF
P2-123	1	Keyed common cam/date shaft	303 stainless, turn/mill/grind	STEP, DXF, PDF
P2-124	2	Cam follower carriage	6061-T6, mill, clear anodize	STEP, DXF, PDF
P2-125	3	Rack clamp	6061-T6, mill, clear anodize	STEP, DXF, PDF
P2-126	1	Carrier output-rack bracket	6061-T6, mill, clear anodize	STEP, DXF, PDF
P2-127	1	Azimuth gear reducer bushing	303 stainless, turn/grind	STEP, DXF, PDF
P2-128	1	Date control shaft	303 stainless, turn/grind	STEP, DXF, PDF

P2-114 is a reference layout, not a separately manufactured part. Do not quote STL files for production; they are inspection/preview derivatives only.

Motion and bearing components

Ref	Qty	Minimum specification	Notes
B01	2 + 1 spare	Track roller, 16 OD x 5 bore x approximately 7 wide, sealed	cam followers
B02	8	Flanged bearing, 8 ID, radial play C2 or standard	display shafts
B03	2	Flanged bushing, 14 ID, dry-running polymer	outer azimuth tube
G01	3	Module 1 rack, 20 degree PA, quality 8+, 120 mm usable teeth	two inputs plus carrier output

Ref	Qty	Minimum specification	Notes
G02	1	Module 1, 30 tooth spur pinion, 8 mm bore	sunrise input
G03	2 + 1 spare	Module 1, 15 tooth spur pinion, 8 mm bore	differential/output
G04	1	Module 1, 80 tooth spur gear, 8 mm bore	sunrise azimuth
G05	1 set	1:1 miter gears, 20 degree PA, 8 mm bores	azimuth reversal
L01	4	100 to 120 mm miniature profile rails with two carriages each	opposed racks/carrier
S01	2	Extension spring, 1 to 2 N installed load, stainless	rack anti-backlash
S02	2	Extension spring, 1 to 3 N over follower travel	cam preload
W01	4	Wave spring washer for 8/14 mm coaxial stack	axial preload
D01	1	HTD-5M timing belt, 15 wide, 760 pitch length	date input, 1:1
D02	2	HTD-5M pulley, 30 tooth, 15 belt width; 12 and 8 mm bores	cam shaft / control shaft
D03	1	Smooth-back timing-belt idler with slotted mount	set belt tension, 10-20 N static

Purchased gears may use imperial bores only if replaceable bushings preserve the specified shaft journals and phase adjustments. Supplier part numbers are left open for the vendor-selection stage.

Shaft and bearing schedule

Shaft	Journal / length	Supports	Mounted elements	Axial retention
P2-123 common cam/date	12 h6 x 160, 4 mm keyway over 70	two 12 mm flanged bearings, 110 span	two P2-103 hubs, D02 pulley	two 12 mm split collars
P2-128 date control	8 h6 x 100	two 8 mm flanged bearings, 55 span	date knob/hand, D02 pulley	shoulder plus 8 mm collar
P2-110 sunrise time	8 h6 x 110	two 8 mm flanged bearings, 65 span	G02 pinion, sunrise hand, input knob	shoulder plus 8 mm collar
P2-111 sunset output	8 h6 x 95	two 8 mm flanged bearings, 55 span	G03 output pinion, sunset hand	shoulder plus 8 mm collar
P2-112 azimuth inner	8 h6 x 150	two 8 mm flanged bearings, 90 span	G04 gear/bushing, sunrise hand	shoulder plus 8 mm collar
P2-113 azimuth tube	14 h8 / 8.10 ID x 130	two 14 mm bushings, 75 span	G05 driven miter, sunset hand	two wave washers and 14 mm collar

Fastener torque schedule

Joint	Fastener	Dry assembly torque	Retention
POM cam to hub	M5 class 8.8 with washer	3 N m maximum	removable threadlocker
Aluminum module plates to T-slot	M6 class 8.8	8 N m	prevailing nut / T-nut
Rack clamps	M4 class 8.8	2.5 N m	removable threadlocker
Follower shoulder screws	M5 shoulder screw	4 N m at thread	prevailing nut
Guard standoffs	M6, nylon washer	3 N m	prevailing nut

Verify that the POM remains flat after torque. Split collars and gear set screws use their supplier torque, never the generic table.

Frame and general hardware

Ref	Qty	Specification
F01	2	20 x 40 T-slot extrusion, 900 long
F02	4	20 x 40 T-slot extrusion, 569 long; two perimeter, two crossmembers
F03	8	20-series right-angle corner brackets with M5 fasteners
F04	12	M6 adjustable standoffs, 35 to 55 long
H01	50	M5 socket head screws, mixed 12/16/20 lengths, class 8.8
H02	30	M6 socket head screws and T-nuts, class 8.8
H03	8	M5 shoulder screws sized for B01 rollers

Ref	Qty	Specification
H04	8	Split shaft collars, 8 mm bore; two with phase slots
H05	2	Split shaft collars, 12 mm bore, keyed/date phase clamp
H06	12	M5 all-metal prevailing torque jam nuts
H07	1 lot	0.1/0.2/0.5 mm stainless shim washers
H08	1 lot	Medium-strength removable threadlocker and witness lacquer

Required inspection equipment (not consumed)

- 0 to 150 mm digital caliper, 0.01 mm display;
- 0 to 25 mm micrometer;
- dial indicator reading 0.01 mm with magnetic or rigid stand;
- 32 mm bore gauge or certified plug/ring masters;
- 4, 5, 6, 8, and 12 mm gauge pins or verified equivalents;
- 0 to 5 N spring scale;
- machinist square and surface plate or verified flat reference;
- protractor or optical angle encoder resolving 0.1 degree;
- printed test record from the included template.

P2 CNC Request-for-Quotation Package

Quote one Rev-A prototype set plus separate unit prices for a second set. Quote P2-120 through P2-122 first as a coupon lot and the remaining custom parts as a production lot released only after coupon approval.

Files and precedence

Use the PDFs for dimensions, GD&T, notes, material, finish, and inspection. Use STEP for 3D form and DXF for flat profiles. CSV cam profiles are independent inspection data. STL is preview-only. All supplied files must match the package SHA-256 list.

Requested quote breakdown

1. POM-C coupon lot: P2-120 and P2-121.
2. POM-C production cams: P2-101 and P2-102.
3. 6061-T6 milled parts: P2-103 through P2-109, P2-122, and P2-124 through P2-126, with anodize shown separately.
4. 303 stainless turned parts: P2-110 through P2-113, P2-123, P2-127, and P2-128.
5. Laser/router parts: P2-115 through P2-119.
6. Inspection report for interfaces I-01 through I-05 and shaft runout.
7. Material certificates, programming/setup charges, shipping, tax, lead time, quote validity, and remake policy as separate entries.

Supplier questions

- Can you hold a 0.05 mm free-form profile relative to a 32 H7 bore in POM-C?
- What stock conditioning and stabilization do you use before final profiling?
- Will the same machine setup finish pilot, datum hole, and profile?
- Can you return measured profile data or a CMM report at 16 date stations?
- What bearing-fit masking method is used during anodize?
- Are any CAD features ambiguous or economically dominant?

No substitution, profile smoothing, unit conversion, or unexplained CAM optimization is authorized. Supplier shall request disposition before deviating.

P2 Vendor Shortlist and Budgetary Estimate

Status: Current-market review, 2026-09-21 This is a sourcing recommendation and planning estimate, not a quotation. No files have been uploaded and no vendor has been contacted or authorized.

Recommended sourcing split

Precision CNC and inspection

Send the same RFQ to Xometry, Protolabs Network, and one local ISO 9001 job shop. Xometry publicly offers 6061, stainless, and acetal with specified tolerances as tight as plus/minus 0.001 inch when called out. Protolabs Network explicitly lists CNC-machined POM-C and provides instant quotation/DFM review. The default platform tolerances are looser than the cam profile requirement, so the 0.05 mm profile, 32 H7/h6 register, stabilization, and inspection report must be treated as custom quote requirements—not platform defaults.

- Xometry CNC capability: <<https://www.xometry.com/capabilities/cnc-machining-service/>>
- Protolabs Network POM capability: <<https://www.hubs.com/cnc-machining/plastic/pom-delrin/>>

Recommendation: release only P2-120 through P2-122 first. Prefer the supplier that accepts the common-datum setup and returns actual measurements, even if it is not the lowest bidder. Then keep the cams and hubs with that same supplier.

Flat guard and hands

SendCutSend is the first-choice benchmark for P2-115, P2-118, and P2-119. Its published materials include 4.5 mm (0.177 inch) CNC-routed polycarbonate and 1.6 mm (0.063 inch) 5052-H32 aluminum. Its published laser/routing tolerance is plus/minus 0.005 inch, suitable for these non-datum flat parts.

- Materials and thicknesses: <<https://sendcutsend.com/materials/>>
- Sheet-cutting tolerances: <<https://sendcutsend.com/services/sheet-cutting/>>

Functional dials

Front Panel Express is the preferred benchmark for P2-116/P2-117 because it combines CNC panel machining with engraving or UV print and live itemized pricing. Use 2.5 mm anodized aluminum; its published capacity at that thickness comfortably includes the 360 mm azimuth dial. Upload the DXF and request only functional markings—decorative artwork is not part of P2.

- Services and live pricing: <<https://www.frontpanelexpress.com/service>>
- Aluminum size capability: <<https://www.frontpanelexpress.com/downloads/FPE/Aluminum-Datasheet.pdf?v2=>>

Gears, racks, and frame

Use KHK USA for the critical module-1 gears: SS1-15 (8 mm bore) for the two 15-tooth elements, SS1-30B (8 mm bore) for sunrise input, SS1-80 with a qualified bushing for azimuth, and an opposite-hand module-1 miter pair. KHK publishes JIS grades and backlash; its ground SSG1-15 was listed at USD 40.48 at review. Use SDP/SI or KHK for module-1, 20-degree racks; SDP/SI listed a 500 mm steel rack at USD 64.64 and a nylon rack at USD 87.96. Cut after receipt and inspection.

- KHK 15-tooth gear: <<https://www.khkgears.us/catalog/product/SSG1-15>>
- KHK module-1 spur catalog: <<https://www.khkgears.us/media/1019/01-spur.pdf>>
- KHK module-1 miter catalog: <<https://khkgears.net/pdf/2025/miter-gears.pdf>>
- SDP/SI steel rack: <<https://shop.sdp-si.com/ksrf1-500.html>>

Use 80/20 20-2040 or a dimensionally equivalent metric profile for the frame. The official profile is 20 x 40 mm 6063-T6, clear anodized, with six slots and published CAD.

- 80/20 20-2040: <<https://8020.net/20-2040.html>>

Budgetary cost range for one prototype

Package	Expected low	Expected high	Basis / main uncertainty
Coupon lot P2-120..122	USD 300	USD 800	inspection/setup dominates
POM-C cams and two hubs	900	2,200	0.05 profile, stabilization, CMM report
Aluminum plates/levers/carrier	700	1,800	setup and anodize
Four stainless shafts/tube	350	900	journals, concentricity, low quantity
Guard and four hands	180	450	routing/laser, deburr, shipping
Three functional dials	550	1,200	engraving/UV print and large diameter
Gears, racks, rails, bearings, springs	750	1,600	gear grade and linear-rail choice
Frame and general hardware	250	550	cut service and fasteners

Package	Expected low	Expected high	Basis / main uncertainty
Shipping, inspection, remake reserve	600	1,500	multi-vendor freight and first article risk
Planning total	4,580	11,000	before tax; no assembly labor

A reasonable authorization budget is USD 7,500, with only USD 300-800 released for the coupon gate initially. The low end assumes automated online quoting and mostly standard inspection; the high end assumes domestic quick-turn work, formal inspection, and one correction cycle. Consolidating all precision parts at one inspected shop may cost more initially but reduces datum-transfer and remake risk.

Route comparison

Route	Cam / precision strategy	Expected total	Planning lead time	Tradeoff
Minimum-cost hybrid	CNC POM-C cams with standard inspection; online-cut guard/hands; commercial-grade racks/gears	USD 4,600-6,500	3-6 weeks after coupon approval	Highest dependence on incoming inspection and preload tuning; remake reserve essential
Recommended accuracy	Stabilized POM-C cams/hubs in one setup with 16-station CMM report; ground 15T gears; inspected rails; anodized datum plates	USD 7,500-11,000	5-10 weeks after coupon approval	Best balance of low cam friction, inspectability, and one-off precision
Aluminum-cam alternate	6061-T6 cams, hard anodized after profile verification, metal rollers	add USD 1,500-3,000	add 2-4 weeks	More dimensionally stable during machining, but greater inertia/noise and anodize thickness must be compensated; not automatically more accurate

POM-C remains the release material because its low friction, machinability, and low moisture uptake suit the lightly loaded follower. POM-H/Delrin is easier to source but is a controlled substitution because dimensional behavior differs. Aluminum is a contingency when a supplier will not certify the 0.05 mm POM-C profile. SLS/MJF nylon, SLA resin, laser-cut stacked cams, and consumer printing are not precision production alternatives for P2; they are useful only for fit/mock-up parts and cannot satisfy the profile/runout allocation without new qualification.

Lead times are planning ranges inferred from the reviewed suppliers' published instant-quote/stock lead-time model and low-volume prototype practice; actual lead time begins only after drawing review and coupon acceptance. Shipping, tax, import duty, and assembly labor are excluded. Before routing live quotes, confirm delivery country/postal code, quantity, budget ceiling, and required in-hand date.

Quote comparison rules

1. Normalize shipping, tax, finish, inspection, and setup before comparing.
2. Reject substitutions from POM-C to POM-H unless thermal/moisture behavior is reviewed and Decision 0005 is revised.
3. Reject default-tolerance quotes that do not acknowledge the critical drawing notes and interface table.
4. Score dimensional evidence, communication, and remake policy above lead time.
5. Do not release production cams until the coupon report is reviewed.

P2 Manufacturing, Assembly, Calibration, and Test Procedure

Status: Release candidate, Rev A This procedure is mandatory evidence for physical acceptance. A clean software verification does not replace it.

1. Incoming file and material control

1. Confirm the quotation references Rev A, millimetres, the PDF drawings, STEP models, DXFs, and the matching SHA256SUMS.txt.
2. Reject a quote that proposes machining from STL.
3. Obtain material certifications for POM-C, 6061-T6, and 303 stainless.
4. Require the machinist to identify any conflict between PDF notes and CAD before cutting. PDF controls tolerances/finish; STEP/DXF controls form.

2. Coupon gate before production cams

1. Machine P2-121 from the same POM-C stock and process intended for the cams.
2. Measure all four marked bores: 31.98, 32.00, 32.02, and 32.04 nominal.
3. Fit the first P2-103 hub. Select the process offset that produces free hand assembly with no detectable rocking; record the actual bore and hub diameter.
4. Machine P2-120 and mount it on that hub with the production screws and dowel.
5. Indicate the 120 mm diameter while rotating the date shaft. Total indicated runout must be 0.05 mm or less. If it fails, isolate shaft, hub-pilot, and profile contributions before authorizing production cams.
6. Inspect P2-122 hole-center distance. It must be 100.00 plus/minus 0.03. Use it to validate the setup used for lever slot and pivot locations.

3. Custom-part inspection

1. Verify manifest hashes and count every part against the BOM.
2. Inspect controlled interfaces I-01 through I-14 from the ICD.
3. On each cam, record thickness at four quadrants, pilot size, datum-hole position, face parallelism, and profile deviation at 16 specified dates.
4. Indicate each shaft journal and hand seat. Total indicated runout must not exceed 0.025 mm.
5. Roll-test purchased racks/gears over their full used travel. Reject tight spots, burrs, visible tooth damage, or backlash above 0.10 mm at the pitch line before spring preload.

4. Cam module assembly

1. Install bearings and the 12 mm date shaft without cams. Confirm free rotation and no more than 0.025 mm radial shaft runout at each hub position.
2. Install P2-128, the two 30-tooth pulleys, 760 mm belt, and idler. Set static belt tension to 10-20 N and verify 1:1 phase over two control revolutions.
3. Install hubs, dowels, and cams with both January 1 rays aligned. Tighten M5 cam screws progressively to 3 N m; POM must not dish or gap from the hub.
4. Install each 16 mm track roller and set 1 to 3 N follower preload. Sweep one full revolution by hand. Contact must be continuous and torque smooth.
5. Record roller-center displacement at the 16 dates in data/p1_test_points_42p1N_2025.csv. Correct for the P2 hub datum only; do not alter the generated profile.

Acceptance: daylight displacement error at most 0.10 mm, azimuth displacement error at most 0.20 mm, repeatability 0.05 mm, and clockwise/counter-clockwise hysteresis 0.10 mm or less. Date control-to-cam phase hysteresis shall be no more than 0.20 degree. These module limits reserve error for transmissions.

5. Time module assembly and calibration

1. Install both opposed racks, rails, 15-tooth differential pinion, carrier, 30-tooth sunrise pinion, and 15-tooth output pinion dry.
2. Align pitch lines within 0.05 mm and install 1 to 2 N anti-backlash springs.
3. Use P2-122 to set the daylight lever nominal ratio to 1.000000. Leave jam nuts finger-tight.
4. With daylight follower at its 545-minute datum, set the sunset phase collar to the same indicated time as sunrise plus 9 h 05 min modulo 12 hours.
5. At winter and summer daylight extrema, adjust lever span equally about the datum until both errors have equal magnitude and opposite sign. Lock nuts, apply witness lacquer, and repeat.
6. Exercise sunrise in 15-minute steps through 12 hours at both daylight extrema. Compare sunset to $(\text{sunrise} + \text{daylight}) \bmod 720$ minutes.

Acceptance: maximum sunset indication error 1.0 minute, repeatability 0.5 minute, and direction-dependent hysteresis 0.5 minute.

6. Azimuth module assembly and calibration

1. Install the 80-tooth pinion, coaxial shaft/tube, 1:1 miter reverser, wave washers, and hands. Preload without perceptible axial play or binding.
2. Set the azimuth lever nominal ratio to 1.000000 using P2-122.
3. At the spring-equinox reference, set the sunrise hand to the reference table and set sunset equal and opposite about north-south.
4. At both solstices, trim lever span for equal and opposite sunrise errors.
5. Sweep all 16 test dates in both directions, reading both hands.

Acceptance: each hand within 1.0 degree of the reference, repeatability 0.5 degree, hysteresis 0.5 degree, and mirror error 1.0 degree.

7. Frame, dial, and guard assembly

1. Square the extrusion frame so diagonal lengths differ by no more than 1 mm.
2. Mount module plates at the ICD Z stations. Verify shafts are normal to the dial plane within 0.1 degree.
3. Install functional dials at the P2-114 centers. Verify every hand clears its dial by 1.5 to 3 mm and adjacent hands by at least 1 mm.
4. Install the guard. Confirm at least 8 mm clearance to every moving part and that routine zero/span fasteners remain reachable after removing only the guard—not another calibrated module.
5. Measure the completed assembly. It must not exceed 609 x 900 x 200.

8. Integrated acceptance sweep

Test the 16 prescribed dates with sunrise settings of 04:00, 06:00, 08:00, and 10:00. Approach every setting clockwise first, then counter-clockwise. Record:

- measured daylight follower displacement;
- sunrise and sunset time indication;
- sunrise and sunset azimuth indication;
- setting direction, ambient temperature, and anomalies.

Pass only when all 64 states meet the time and azimuth acceptance limits and no follower loses contact. Photograph the mechanism at both solstices and archive the completed build record, raw data, deviations, and disposition.

9. Stop conditions

Stop and investigate rather than re-machining a generated profile when runout exceeds the coupon limit, the follower unloads, a rack binds, the guard is contacted, calibration requires more than the documented adjustment range, or results depend on approach direction beyond the hysteresis limit.